

Create a Database

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!pip install -U datasets transformers sentence-transformers
huggingface-hub
!pip install -U langchain langchain-community langchain-classic
langchain-text-splitters
!pip install -U langchain-huggingface langchain-chroma chromadb

Requirement already satisfied: datasets in
/usr/local/lib/python3.13/dist-packages (5.0.1)
Requirement already satisfied: transformers in
/usr/local/lib/python3.13/dist-packages (5.17.0)
Requirement already satisfied: sentence-transformers in
/usr/local/lib/python3.13/dist-packages (6.1.0)
Requirement already satisfied: huggingface-hub in
/usr/local/lib/python3.13/dist-packages (1.32.0)
Requirement already satisfied: filelock in
/usr/local/lib/python3.13/dist-packages (from datasets) (3.32.5)
Requirement already satisfied: numpy>=1.17 in
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Requirement already satisfied: dill<0.4.2,>=0.3.0 in
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Requirement already satisfied: pandas in
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Requirement already satisfied: requests>=2.32.2 in
/usr/local/lib/python3.13/dist-packages (from datasets) (2.34.2)
Requirement already satisfied: httpx<1.0.0 in
/usr/local/lib/python3.13/dist-packages (from datasets) (0.28.1)
Requirement already satisfied: tqdm>=4.66.3 in
/usr/local/lib/python3.13/dist-packages (from datasets) (4.67.3)
Requirement already satisfied: xxhash in
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Requirement already satisfied: pyyaml>=5.1 in
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Requirement already satisfied: regex>=2025.10.22 in
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(2025.11.3)
Requirement already satisfied: tokenizers<0.24.0,>=0.23.1 in
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(2.11.0+cu128)
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(1.16.3)
Requirement already satisfied: typing_extensions>=4.10.0 in
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(4.16.0)
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(2026.7.22)
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(1.0.9)
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Requirement already satisfied: threadpoolctl>=3.1.0 in
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Requirement already satisfied: anyio>=3.5.0 in
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Requirement already satisfied: python-dotenv>=0.21.0 in
/usr/local/lib/python3.13/dist-packages (from pydantic-
settings<3.0.0,>=2.10.1->langchain-community) (1.2.3)
Requirement already satisfied: charset_normalizer<4,>=2 in
/usr/local/lib/python3.13/dist-packages (from requests<3.0.0,>=2.32.5-
>langchain-community) (3.4.9)
Requirement already satisfied: idna<4,>=2.5 in
/usr/local/lib/python3.13/dist-packages (from requests<3.0.0,>=2.32.5-
>langchain-community) (3.19)
Requirement already satisfied: urllib3<3,>=1.26 in
/usr/local/lib/python3.13/dist-packages (from requests<3.0.0,>=2.32.5-
>langchain-community) (2.5.0)
Requirement already satisfied: certifi>=2023.5.7 in
/usr/local/lib/python3.13/dist-packages (from requests<3.0.0,>=2.32.5-
>langchain-community) (2026.7.22)

Requirement already satisfied: greenlet>=1 in
/usr/local/lib/python3.13/dist-packages (from
sqlalchemy<3.0.0,>=1.4.0->langchain-community) (3.5.5)

Requirement already satisfied: httpcore==1.* in
/usr/local/lib/python3.13/dist-packages (from httpx<1.0.0,>=0.23.0->langchain-core<2.0.0,>=1.6.3->langchain) (1.0.9)

Requirement already satisfied: h11>=0.16 in
/usr/local/lib/python3.13/dist-packages (from httpcore==1.*->httpx<1.0.0,>=0.23.0->langchain-core<2.0.0,>=1.6.3->langchain) (0.16.0)

Requirement already satisfied: httpcore2==2.12.0 in
/usr/local/lib/python3.13/dist-packages (from httpx2<3,>=2->langsmith<1.0.0,>=0.1.125->langchain-community) (2.12.0)

Requirement already satisfied: truststore>=0.10 in
/usr/local/lib/python3.13/dist-packages (from httpx2<3,>=2->langsmith<1.0.0,>=0.1.125->langchain-community) (0.10.4)

Requirement already satisfied: jsonpointer>=1.9 in
/usr/local/lib/python3.13/dist-packages (from
jsonpatch<2.0.0,>=1.33.0->langchain-core<2.0.0,>=1.6.3->langchain) (3.1.1)

Requirement already satisfied: ormsgpack>=1.12.0 in
/usr/local/lib/python3.13/dist-packages (from langgraph-checkpoint<5.0.0,>=4.1.0->langgraph<1.3.0,>=1.2.11->langchain) (1.12.2)

Requirement already satisfied: langchain-huggingface in
/usr/local/lib/python3.13/dist-packages (1.2.2)

Requirement already satisfied: langchain-chroma in
/usr/local/lib/python3.13/dist-packages (1.1.0)

Requirement already satisfied: chromadb in
/usr/local/lib/python3.13/dist-packages (1.5.9)

Requirement already satisfied: huggingface-hub<2.0.0,>=0.33.4 in
/usr/local/lib/python3.13/dist-packages (from langchain-huggingface) (1.32.0)

Requirement already satisfied: langchain-core<2.0.0,>=1.2.31 in
/usr/local/lib/python3.13/dist-packages (from langchain-huggingface) (1.6.4)

Requirement already satisfied: tokenizers<1.0.0,>=0.19.1 in
/usr/local/lib/python3.13/dist-packages (from langchain-huggingface) (0.23.1)

Requirement already satisfied: numpy>=2.1.0 in
/usr/local/lib/python3.13/dist-packages (from langchain-chroma) (2.1.3)

Requirement already satisfied: build>=1.0.3 in
/usr/local/lib/python3.13/dist-packages (from chromadb) (1.6.1)

Requirement already satisfied: pydantic>=2.0 in
/usr/local/lib/python3.13/dist-packages (from chromadb) (2.13.5)

Requirement already satisfied: pydantic-settings>=2.0 in
/usr/local/lib/python3.13/dist-packages (from chromadb) (2.15.0)

Requirement already satisfied: pybase64>=1.4.1 in

/usr/local/lib/python3.13/dist-packages (from chromadb) (1.5.0)
Requirement already satisfied: uvicorn>=0.18.3 in
/usr/local/lib/python3.13/dist-packages (from
uvicorn[standard]>=0.18.3->chromadb) (0.52.4)
Requirement already satisfied: typing-extensions>=4.5.0 in
/usr/local/lib/python3.13/dist-packages (from chromadb) (4.16.0)
Requirement already satisfied: onnxruntime>=1.14.1 in
/usr/local/lib/python3.13/dist-packages (from chromadb) (1.30.0)
Requirement already satisfied: opentelemetry-api>=1.2.0 in
/usr/local/lib/python3.13/dist-packages (from chromadb) (1.44.0)
Requirement already satisfied: opentelemetry-exporter-otlp-proto-
grpc>=1.2.0 in /usr/local/lib/python3.13/dist-packages (from chromadb)
(1.44.0)
Requirement already satisfied: opentelemetry-sdk>=1.2.0 in
/usr/local/lib/python3.13/dist-packages (from chromadb) (1.44.0)
Requirement already satisfied: pypika>=0.48.9 in
/usr/local/lib/python3.13/dist-packages (from chromadb) (0.51.1)
Requirement already satisfied: tqdm>=4.65.0 in
/usr/local/lib/python3.13/dist-packages (from chromadb) (4.67.3)
Requirement already satisfied: overrides>=7.3.1 in
/usr/local/lib/python3.13/dist-packages (from chromadb) (7.7.0)
Requirement already satisfied: importlib-resources in
/usr/local/lib/python3.13/dist-packages (from chromadb) (7.1.0)
Requirement already satisfied: grpcio>=1.58.0 in
/usr/local/lib/python3.13/dist-packages (from chromadb) (1.83.1)
Requirement already satisfied: bcrypt>=4.0.1 in
/usr/local/lib/python3.13/dist-packages (from chromadb) (5.0.0)
Requirement already satisfied: typer>=0.9.0 in
/usr/local/lib/python3.13/dist-packages (from chromadb) (0.27.2)
Requirement already satisfied: kubernetes>=28.1.0 in
/usr/local/lib/python3.13/dist-packages (from chromadb) (36.0.3)
Requirement already satisfied: tenacity>=8.2.3 in
/usr/local/lib/python3.13/dist-packages (from chromadb) (9.1.4)
Requirement already satisfied: pyyaml>=6.0.0 in
/usr/local/lib/python3.13/dist-packages (from chromadb) (6.0.3)
Requirement already satisfied: mmh3>=4.0.1 in
/usr/local/lib/python3.13/dist-packages (from chromadb) (5.3.0)
Requirement already satisfied: orjson>=3.9.12 in
/usr/local/lib/python3.13/dist-packages (from chromadb) (3.12.0)
Requirement already satisfied: httpx>=0.27.0 in
/usr/local/lib/python3.13/dist-packages (from chromadb) (0.28.1)
Requirement already satisfied: rich>=10.11.0 in
/usr/local/lib/python3.13/dist-packages (from chromadb) (13.9.4)
Requirement already satisfied: jsonschema>=4.19.0 in
/usr/local/lib/python3.13/dist-packages (from chromadb) (4.26.0)
Requirement already satisfied: packaging>=24.0 in
/usr/local/lib/python3.13/dist-packages (from build>=1.0.3->chromadb)
(26.3)
Requirement already satisfied: pyproject_hooks in

/usr/local/lib/python3.13/dist-packages (from build>=1.0.3->chromadb) (1.3.3)
Requirement already satisfied: anyio in /usr/local/lib/python3.13/dist-packages (from httpx>=0.27.0->chromadb) (4.14.2)
Requirement already satisfied: certifi in /usr/local/lib/python3.13/dist-packages (from httpx>=0.27.0->chromadb) (2026.7.22)
Requirement already satisfied: httpcore==1.* in /usr/local/lib/python3.13/dist-packages (from httpx>=0.27.0->chromadb) (1.0.9)
Requirement already satisfied: idna in /usr/local/lib/python3.13/dist-packages (from httpx>=0.27.0->chromadb) (3.19)
Requirement already satisfied: h11>=0.16 in /usr/local/lib/python3.13/dist-packages (from httpcore==1.*->httpx>=0.27.0->chromadb) (0.16.0)
Requirement already satisfied: click<9.0.0,>=8.4.2 in /usr/local/lib/python3.13/dist-packages (from huggingface-hub<2.0.0,>=0.33.4->langchain-huggingface) (8.5.0)
Requirement already satisfied: filelock>=3.10.0 in /usr/local/lib/python3.13/dist-packages (from huggingface-hub<2.0.0,>=0.33.4->langchain-huggingface) (3.32.5)
Requirement already satisfied: fsspec>=2023.5.0 in /usr/local/lib/python3.13/dist-packages (from huggingface-hub<2.0.0,>=0.33.4->langchain-huggingface) (2025.12.0)
Requirement already satisfied: hf-xet<2.0.0,>=1.5.2 in /usr/local/lib/python3.13/dist-packages (from huggingface-hub<2.0.0,>=0.33.4->langchain-huggingface) (1.6.0)
Requirement already satisfied: attrs>=22.2.0 in /usr/local/lib/python3.13/dist-packages (from jsonschema>=4.19.0->chromadb) (26.1.0)
Requirement already satisfied: jsonschema-specifications>=2023.03.6 in /usr/local/lib/python3.13/dist-packages (from jsonschema>=4.19.0->chromadb) (2025.9.1)
Requirement already satisfied: referencing>=0.28.4 in /usr/local/lib/python3.13/dist-packages (from jsonschema>=4.19.0->chromadb) (0.37.0)
Requirement already satisfied: rpds-py>=0.25.0 in /usr/local/lib/python3.13/dist-packages (from jsonschema>=4.19.0->chromadb) (2026.6.3)
Requirement already satisfied: six>=1.9.0 in /usr/local/lib/python3.13/dist-packages (from kubernetes>=28.1.0->chromadb) (1.17.0)
Requirement already satisfied: python-dateutil>=2.5.3 in /usr/local/lib/python3.13/dist-packages (from kubernetes>=28.1.0->chromadb) (2.9.0.post0)
Requirement already satisfied: websocket-client!=0.40.0,!0.41.*,!0.42.*,>=0.32.0 in /usr/local/lib/python3.13/dist-packages (from kubernetes>=28.1.0->chromadb) (1.9.2)
Requirement already satisfied: requests in

/usr/local/lib/python3.13/dist-packages (from kubernetes>=28.1.0->chromadb) (2.34.2)
Requirement already satisfied: requests-oauthlib in /usr/local/lib/python3.13/dist-packages (from kubernetes>=28.1.0->chromadb) (2.0.0)
Requirement already satisfied: urllib3!=2.6.0,>=1.24.2 in /usr/local/lib/python3.13/dist-packages (from kubernetes>=28.1.0->chromadb) (2.5.0)
Requirement already satisfied: durationpy>=0.7 in /usr/local/lib/python3.13/dist-packages (from kubernetes>=28.1.0->chromadb) (0.11)
Requirement already satisfied: aiohttp<4.0.0,>=3.13.5 in /usr/local/lib/python3.13/dist-packages (from kubernetes>=28.1.0->chromadb) (3.14.3)
Requirement already satisfied: jsonpatch<2.0.0,>=1.33.0 in /usr/local/lib/python3.13/dist-packages (from langchain-core<2.0.0,>=1.2.31->langchain-huggingface) (1.33)
Requirement already satisfied: langchain-protocol>=0.0.17 in /usr/local/lib/python3.13/dist-packages (from langchain-core<2.0.0,>=1.2.31->langchain-huggingface) (0.0.19)
Requirement already satisfied: langsmith<1.0.0,>=0.3.45 in /usr/local/lib/python3.13/dist-packages (from langchain-core<2.0.0,>=1.2.31->langchain-huggingface) (0.12.1)
Requirement already satisfied: uuid-utils<1.0,>=0.12.0 in /usr/local/lib/python3.13/dist-packages (from langchain-core<2.0.0,>=1.2.31->langchain-huggingface) (0.17.0)
Requirement already satisfied: flatbuffers in /usr/local/lib/python3.13/dist-packages (from onnxruntime>=1.14.1->chromadb) (25.12.19)
Requirement already satisfied: protobuf>=4.25.8 in /usr/local/lib/python3.13/dist-packages (from onnxruntime>=1.14.1->chromadb) (5.29.6)
Requirement already satisfied: googleapis-common-protos~=1.57 in /usr/local/lib/python3.13/dist-packages (from opentelemetry-exporter-otlp-proto-grpc>=1.2.0->chromadb) (1.75.0)
Requirement already satisfied: opentelemetry-exporter-otlp-proto-common==1.44.0 in /usr/local/lib/python3.13/dist-packages (from opentelemetry-exporter-otlp-proto-grpc>=1.2.0->chromadb) (1.44.0)
Requirement already satisfied: opentelemetry-proto==1.44.0 in /usr/local/lib/python3.13/dist-packages (from opentelemetry-exporter-otlp-proto-grpc>=1.2.0->chromadb) (1.44.0)
Requirement already satisfied: opentelemetry-semantic-conventions==0.65b0 in /usr/local/lib/python3.13/dist-packages (from opentelemetry-sdk>=1.2.0->chromadb) (0.65b0)
Requirement already satisfied: annotated-types>=0.6.0 in /usr/local/lib/python3.13/dist-packages (from pydantic>=2.0->chromadb) (0.8.0)
Requirement already satisfied: pydantic-core==2.46.5 in /usr/local/lib/python3.13/dist-packages (from pydantic>=2.0->chromadb)

(2.46.5)

Requirement already satisfied: typing-inspection>=0.4.2 in
/usr/local/lib/python3.13/dist-packages (from pydantic>=2.0->chromadb)
(0.4.4)

Requirement already satisfied: python-dotenv>=0.21.0 in
/usr/local/lib/python3.13/dist-packages (from pydantic-settings>=2.0->chromadb) (1.2.3)

Requirement already satisfied: markdown-it-py>=2.2.0 in
/usr/local/lib/python3.13/dist-packages (from rich>=10.11.0->chromadb)
(4.2.0)

Requirement already satisfied: pygments<3.0.0,>=2.13.0 in
/usr/local/lib/python3.13/dist-packages (from rich>=10.11.0->chromadb)
(2.21.0)

Requirement already satisfied: shellingham>=1.3.0 in
/usr/local/lib/python3.13/dist-packages (from typer>=0.9.0->chromadb)
(1.5.4)

Requirement already satisfied: annotated-doc>=0.0.2 in
/usr/local/lib/python3.13/dist-packages (from typer>=0.9.0->chromadb)
(0.0.5)

Requirement already satisfied: httptools>=0.8.0 in
/usr/local/lib/python3.13/dist-packages (from
uvicorn[standard]>=0.18.3->chromadb) (0.8.0)

Requirement already satisfied: uvloop>=0.15.1 in
/usr/local/lib/python3.13/dist-packages (from
uvicorn[standard]>=0.18.3->chromadb) (0.22.1)

Requirement already satisfied: watchfiles>=0.20 in
/usr/local/lib/python3.13/dist-packages (from
uvicorn[standard]>=0.18.3->chromadb) (1.2.0)

Requirement already satisfied: websockets>=13.0 in
/usr/local/lib/python3.13/dist-packages (from
uvicorn[standard]>=0.18.3->chromadb) (15.0.1)

Requirement already satisfied: aiohappyeyeballs>=2.5.0 in
/usr/local/lib/python3.13/dist-packages (from aiohttp<4.0.0,>=3.13.5->kubernetes>=28.1.0->chromadb) (2.7.1)

Requirement already satisfied: aiosignal>=1.4.0 in
/usr/local/lib/python3.13/dist-packages (from aiohttp<4.0.0,>=3.13.5->kubernetes>=28.1.0->chromadb) (1.4.0)

Requirement already satisfied: frozenlist>=1.1.1 in
/usr/local/lib/python3.13/dist-packages (from aiohttp<4.0.0,>=3.13.5->kubernetes>=28.1.0->chromadb) (1.8.0)

Requirement already satisfied: multidict<7.0,>=4.5 in
/usr/local/lib/python3.13/dist-packages (from aiohttp<4.0.0,>=3.13.5->kubernetes>=28.1.0->chromadb) (6.7.1)

Requirement already satisfied: propcache>=0.2.0 in
/usr/local/lib/python3.13/dist-packages (from aiohttp<4.0.0,>=3.13.5->kubernetes>=28.1.0->chromadb) (0.5.2)

Requirement already satisfied: yarl<2.0,>=1.17.0 in
/usr/local/lib/python3.13/dist-packages (from aiohttp<4.0.0,>=3.13.5->kubernetes>=28.1.0->chromadb) (1.24.5)

Requirement already satisfied: jsonpointer>=1.9 in
/usr/local/lib/python3.13/dist-packages (from
jsonpatch<2.0.0,>=1.33.0->langchain-core<2.0.0,>=1.2.31->langchain-
huggingface) (3.1.1)

Requirement already satisfied: distro>=1.7.0 in
/usr/local/lib/python3.13/dist-packages (from
langsmith<1.0.0,>=0.3.45->langchain-core<2.0.0,>=1.2.31->langchain-
huggingface) (1.9.0)

Requirement already satisfied: httpx2<3,>=2 in
/usr/local/lib/python3.13/dist-packages (from
langsmith<1.0.0,>=0.3.45->langchain-core<2.0.0,>=1.2.31->langchain-
huggingface) (2.12.0)

Requirement already satisfied: requests-toolbelt>=1.0.0 in
/usr/local/lib/python3.13/dist-packages (from
langsmith<1.0.0,>=0.3.45->langchain-core<2.0.0,>=1.2.31->langchain-
huggingface) (1.0.0)

Requirement already satisfied: sniffio>=1.1 in
/usr/local/lib/python3.13/dist-packages (from
langsmith<1.0.0,>=0.3.45->langchain-core<2.0.0,>=1.2.31->langchain-
huggingface) (1.3.1)

Requirement already satisfied: xxhash>=3.0.0 in
/usr/local/lib/python3.13/dist-packages (from
langsmith<1.0.0,>=0.3.45->langchain-core<2.0.0,>=1.2.31->langchain-
huggingface) (4.0.1)

Requirement already satisfied: zstandard>=0.23.0 in
/usr/local/lib/python3.13/dist-packages (from
langsmith<1.0.0,>=0.3.45->langchain-core<2.0.0,>=1.2.31->langchain-
huggingface) (0.25.0)

Requirement already satisfied: mdurl~=0.1 in
/usr/local/lib/python3.13/dist-packages (from markdown-it-py>=2.2.0-
>rich>=10.11.0->chromadb) (0.1.2)

Requirement already satisfied: charset_normalizer<4,>=2 in
/usr/local/lib/python3.13/dist-packages (from requests-
>kubernetes>=28.1.0->chromadb) (3.4.9)

Requirement already satisfied: oauthlib>=3.0.0 in
/usr/local/lib/python3.13/dist-packages (from requests-oauthlib-
>kubernetes>=28.1.0->chromadb) (3.3.1)

Requirement already satisfied: httpcore2==2.12.0 in
/usr/local/lib/python3.13/dist-packages (from httpx2<3,>=2-
>langsmith<1.0.0,>=0.3.45->langchain-core<2.0.0,>=1.2.31->langchain-
huggingface) (2.12.0)

Requirement already satisfied: truststore>=0.10 in
/usr/local/lib/python3.13/dist-packages (from httpx2<3,>=2-
>langsmith<1.0.0,>=0.3.45->langchain-core<2.0.0,>=1.2.31->langchain-
huggingface) (0.10.4)

```
import os
import shutil
from transformers import AutoModelForCausalLM, BitsAndBytesConfig,
AutoTokenizer, pipeline
```

```

import torch

from langchain_classic.chains import RetrievalQA
from langchain_community.document_loaders import DirectoryLoader,
TextLoader
from langchain_text_splitters import RecursiveCharacterTextSplitter
from langchain_core.documents import Document
from langchain_huggingface import HuggingFaceEmbeddings,
HuggingFacePipeline
from langchain_chroma import Chroma

/tmp/ipykernel_4671/1806037342.py:7: DeprecationWarning: `langchain-
community` is being sunset and is no longer actively maintained. See
https://github.com/langchain-ai/langchain-community/issues/674 for
details and migration guidance toward standalone integration packages.
  from langchain_community.document_loaders import DirectoryLoader,
TextLoader

from google.colab import drive
drive.mount('/content/drive')

Drive already mounted at /content/drive; to attempt to forcibly
remount, call drive.mount("/content/drive", force_remount=True).

CHROMA_PATH = "/content/drive/My Drive/chromadb"
DATA_PATH = "/content/drive/My Drive/RAG_info.txt"

loader = TextLoader(DATA_PATH)
documents = loader.load()

def split_text(documents: list[Document]):
    text_splitter = RecursiveCharacterTextSplitter(
        chunk_size=300,
        chunk_overlap=100,
        length_function=len,
        add_start_index=True,
    )
    chunks = text_splitter.split_documents(documents)
    print(f"Split {len(documents)} documents into {len(chunks)}
chunks.")

    document = chunks[min(10, len(chunks) - 1)]
    print(document.page_content)
    print(document.metadata)

    return chunks

chunks = split_text(documents)

if os.path.exists(CHROMA_PATH):

```

```

shutil.rmtree(CHROMA_PATH)

embeddings =
HuggingFaceEmbeddings(model_name="sentence-transformers/all-MiniLM-L6-
v2")

# Create a new DB from the documents.
db = Chroma.from_documents(
    chunks, embeddings, ids=None, collection_name="langchain",
    persist_directory= CHROMA_PATH
)

# The database is written to CHROMA_PATH automatically (db.persist()
no longer exists)
print(f"Saved {len(chunks)} chunks to {CHROMA_PATH}.")

Split 1 documents into 145 chunks.
needed for growth and development. However, achieving a healthy diet
and regular exercise in childhood can be challenging, given the
prevalence of unhealthy food consumption and sedentary lifestyles
characterized by long periods of screen time. Some of the first large-
scale studies associating
{'source': '/content/drive/My Drive/RAG_info.txt', 'start_index':
2064}

{"model_id": "6de67d4c9a504deaadbba0d50df283b9", "version_major": 2, "vers
ion_minor": 0}

Saved 145 chunks to /content/drive/My Drive/chromadb.

query = "How can I manage stress and maintain a healthy work-life
balance?"
docs = db.similarity_search(query, k=1)

# print results
print(docs[0].page_content)

limiting caffeine and alcohol consumption
getting enough sleep
getting regular exercise
meditating
setting aside free time for hobbies and activities that bring you joy
keeping a diary of your feelings and things that trigger stress
practicing breathing exercises

retriever = db.as_retriever()

query = "How can I manage stress and maintain a healthy work-life
balance?"

# Retrieve the most relevant documents (.invoke replaces

```

```

get_relevant_documents)
retrieved_docs = retriever.invoke(query)

# Print the retrieved documents
for doc in retrieved_docs:
    print(doc)

page_content='limiting caffeine and alcohol consumption
getting enough sleep
getting regular exercise
meditating
setting aside free time for hobbies and activities that bring you joy
keeping a diary of your feelings and things that trigger stress
practicing breathing exercises' metadata={'source': '/content/drive/My
Drive/RAG_info.txt', 'start_index': 13705}
page_content='It's common to experience stress and anxiety from time
to time, and there are strategies you can use to make them more
manageable.' metadata={'source': '/content/drive/My
Drive/RAG_info.txt', 'start_index': 13161}
page_content='keeping a diary of your feelings and things that trigger
stress
practicing breathing exercises
opening up to loved ones about how you're feeling' metadata={'source':
'/content/drive/My Drive/RAG_info.txt', 'start_index': 13874}
page_content='Maybe you have an upcoming exam that you're worried
about taking. Or you're trying to juggle working from home with three
small children who are competing for your attention. In both cases,
there's a specific root of your stress. Once the exam is over or your
children return to daycare, your stress' metadata={'source':
'/content/drive/My Drive/RAG_info.txt', 'start_index': 11478}

from huggingface_hub import notebook_login
notebook_login()

!pip install -U bitsandbytes accelerate peft

from transformers import AutoModelForCausalLM
from peft import PeftModel, PeftConfig

# Load the PEFT config
config = PeftConfig.from_pretrained("Sahaj10/my_fine_tuned_llama2")

# Load the base model in 8-bit precision.
base_model = AutoModelForCausalLM.from_pretrained(
    "genaitraining/llama-2-7b-domain-tuned",
    quantization_config=BitsAndBytesConfig(load_in_8bit=True), # This
enables loading in 8-bit precision
    device_map='auto' # This can automatically set the device map
for distributed training

```

)

Load the PEFT model with the fine-tuned weights

```
model = PeftModel.from_pretrained(base_model,  
"Sahaj10/my_fine_tuned_llama2")
```

Requirement already satisfied: bitsandbytes in
/usr/local/lib/python3.13/dist-packages (0.50.2)

Requirement already satisfied: accelerate in
/usr/local/lib/python3.13/dist-packages (1.15.0)

Requirement already satisfied: peft in /usr/local/lib/python3.13/dist-packages (0.21.0)

Requirement already satisfied: torch<3,>=2.4 in
/usr/local/lib/python3.13/dist-packages (from bitsandbytes)
(2.11.0+cu128)

Requirement already satisfied: numpy>=1.17 in
/usr/local/lib/python3.13/dist-packages (from bitsandbytes) (2.1.3)

Requirement already satisfied: packaging>=20.9 in
/usr/local/lib/python3.13/dist-packages (from bitsandbytes) (26.3)

Requirement already satisfied: psutil in
/usr/local/lib/python3.13/dist-packages (from accelerate) (5.9.5)

Requirement already satisfied: pyyaml in
/usr/local/lib/python3.13/dist-packages (from accelerate) (6.0.3)

Requirement already satisfied: huggingface_hub>=0.21.0 in
/usr/local/lib/python3.13/dist-packages (from accelerate) (1.32.0)

Requirement already satisfied: safetensors>=0.4.3 in
/usr/local/lib/python3.13/dist-packages (from accelerate) (0.8.0)

Requirement already satisfied: transformers in
/usr/local/lib/python3.13/dist-packages (from peft) (5.17.0)

Requirement already satisfied: tqdm in /usr/local/lib/python3.13/dist-packages (from peft) (4.67.3)

Requirement already satisfied: click<9.0.0,>=8.4.2 in
/usr/local/lib/python3.13/dist-packages (from huggingface_hub>=0.21.0->accelerate) (8.5.0)

Requirement already satisfied: filelock>=3.10.0 in
/usr/local/lib/python3.13/dist-packages (from huggingface_hub>=0.21.0->accelerate) (3.32.5)

Requirement already satisfied: fsspec>=2023.5.0 in
/usr/local/lib/python3.13/dist-packages (from huggingface_hub>=0.21.0->accelerate) (2025.12.0)

Requirement already satisfied: hf-xet<2.0.0,>=1.5.2 in
/usr/local/lib/python3.13/dist-packages (from huggingface_hub>=0.21.0->accelerate) (1.6.0)

Requirement already satisfied: httpx<1,>=0.23.0 in
/usr/local/lib/python3.13/dist-packages (from huggingface_hub>=0.21.0->accelerate) (0.28.1)

Requirement already satisfied: typing-extensions>=4.1.0 in
/usr/local/lib/python3.13/dist-packages (from huggingface_hub>=0.21.0->accelerate) (4.16.0)

Requirement already satisfied: setuptools<82 in

```
/usr/local/lib/python3.13/dist-packages (from torch<3,>=2.4-
>bitsandbytes) (80.10.2)
Requirement already satisfied: sympy>=1.13.3 in
/usr/local/lib/python3.13/dist-packages (from torch<3,>=2.4-
>bitsandbytes) (1.14.0)
Requirement already satisfied: networkx>=2.5.1 in
/usr/local/lib/python3.13/dist-packages (from torch<3,>=2.4-
>bitsandbytes) (3.6.1)
Requirement already satisfied: jinja2 in
/usr/local/lib/python3.13/dist-packages (from torch<3,>=2.4-
>bitsandbytes) (3.1.6)
Requirement already satisfied: cuda-toolkit==12.8.1 in
/usr/local/lib/python3.13/dist-packages (from cuda-
toolkit[cublas,cudart,cufft,cufile,cupti,curand,cusolver,cusparse,nvji
tlink,nvrtc,nvtx]==12.8.1; platform_system == "Linux"->torch<3,>=2.4-
>bitsandbytes) (12.8.1)
Requirement already satisfied: cuda-bindings<13,>=12.9.4 in
/usr/local/lib/python3.13/dist-packages (from torch<3,>=2.4-
>bitsandbytes) (12.9.7)
Requirement already satisfied: nvidia-cudnn-cu12==9.19.0.56 in
/usr/local/lib/python3.13/dist-packages (from torch<3,>=2.4-
>bitsandbytes) (9.19.0.56)
Requirement already satisfied: nvidia-cusparselt-cu12==0.7.1 in
/usr/local/lib/python3.13/dist-packages (from torch<3,>=2.4-
>bitsandbytes) (0.7.1)
Requirement already satisfied: nvidia-nccl-cu12==2.28.9 in
/usr/local/lib/python3.13/dist-packages (from torch<3,>=2.4-
>bitsandbytes) (2.28.9)
Requirement already satisfied: nvidia-nvshmem-cu12==3.4.5 in
/usr/local/lib/python3.13/dist-packages (from torch<3,>=2.4-
>bitsandbytes) (3.4.5)
Requirement already satisfied: triton==3.6.0 in
/usr/local/lib/python3.13/dist-packages (from torch<3,>=2.4-
>bitsandbytes) (3.6.0)
Requirement already satisfied: nvidia-cublas-cu12==12.8.4.1.* in
/usr/local/lib/python3.13/dist-packages (from cuda-
toolkit[cublas,cudart,cufft,cufile,cupti,curand,cusolver,cusparse,nvji
tlink,nvrtc,nvtx]==12.8.1; platform_system == "Linux"->torch<3,>=2.4-
>bitsandbytes) (12.8.4.1)
Requirement already satisfied: nvidia-cuda-runtime-cu12==12.8.90.*
in /usr/local/lib/python3.13/dist-packages (from cuda-
toolkit[cublas,cudart,cufft,cufile,cupti,curand,cusolver,cusparse,nvji
tlink,nvrtc,nvtx]==12.8.1; platform_system == "Linux"->torch<3,>=2.4-
>bitsandbytes) (12.8.90)
Requirement already satisfied: nvidia-cufft-cu12==11.3.3.83.* in
/usr/local/lib/python3.13/dist-packages (from cuda-
toolkit[cublas,cudart,cufft,cufile,cupti,curand,cusolver,cusparse,nvji
tlink,nvrtc,nvtx]==12.8.1; platform_system == "Linux"->torch<3,>=2.4-
>bitsandbytes) (11.3.3.83)
```

Requirement already satisfied: nvidia-cufile-cu12==1.13.1.3.* in /usr/local/lib/python3.13/dist-packages (from cuda-toolkit[cublas,cudart,cufft,cufile,cupti,curand,cusolver,cusparsenvjitlink,nvrtc,nvtx]==12.8.1; platform_system == "Linux"->torch<3,>=2.4->bitsandbytes) (1.13.1.3)

Requirement already satisfied: nvidia-cuda-cupti-cu12==12.8.90.* in /usr/local/lib/python3.13/dist-packages (from cuda-toolkit[cublas,cudart,cufft,cufile,cupti,curand,cusolver,cusparsenvjitlink,nvrtc,nvtx]==12.8.1; platform_system == "Linux"->torch<3,>=2.4->bitsandbytes) (12.8.90)

Requirement already satisfied: nvidia-curand-cu12==10.3.9.90.* in /usr/local/lib/python3.13/dist-packages (from cuda-toolkit[cublas,cudart,cufft,cufile,cupti,curand,cusolver,cusparsenvjitlink,nvrtc,nvtx]==12.8.1; platform_system == "Linux"->torch<3,>=2.4->bitsandbytes) (10.3.9.90)

Requirement already satisfied: nvidia-cusolver-cu12==11.7.3.90.* in /usr/local/lib/python3.13/dist-packages (from cuda-toolkit[cublas,cudart,cufft,cufile,cupti,curand,cusolver,cusparsenvjitlink,nvrtc,nvtx]==12.8.1; platform_system == "Linux"->torch<3,>=2.4->bitsandbytes) (11.7.3.90)

Requirement already satisfied: nvidia-cusparsenvjitlink-cu12==12.5.8.93.* in /usr/local/lib/python3.13/dist-packages (from cuda-toolkit[cublas,cudart,cufft,cufile,cupti,curand,cusolver,cusparsenvjitlink,nvrtc,nvtx]==12.8.1; platform_system == "Linux"->torch<3,>=2.4->bitsandbytes) (12.5.8.93)

Requirement already satisfied: nvidia-nvjitlink-cu12==12.8.93.* in /usr/local/lib/python3.13/dist-packages (from cuda-toolkit[cublas,cudart,cufft,cufile,cupti,curand,cusolver,cusparsenvjitlink,nvrtc,nvtx]==12.8.1; platform_system == "Linux"->torch<3,>=2.4->bitsandbytes) (12.8.93)

Requirement already satisfied: nvidia-cuda-nvrtc-cu12==12.8.93.* in /usr/local/lib/python3.13/dist-packages (from cuda-toolkit[cublas,cudart,cufft,cufile,cupti,curand,cusolver,cusparsenvjitlink,nvrtc,nvtx]==12.8.1; platform_system == "Linux"->torch<3,>=2.4->bitsandbytes) (12.8.93)

Requirement already satisfied: nvidia-nvtx-cu12==12.8.90.* in /usr/local/lib/python3.13/dist-packages (from cuda-toolkit[cublas,cudart,cufft,cufile,cupti,curand,cusolver,cusparsenvjitlink,nvrtc,nvtx]==12.8.1; platform_system == "Linux"->torch<3,>=2.4->bitsandbytes) (12.8.90)

Requirement already satisfied: regex>=2025.10.22 in /usr/local/lib/python3.13/dist-packages (from transformers->peft) (2025.11.3)

Requirement already satisfied: tokenizers<0.24.0,>=0.23.1 in /usr/local/lib/python3.13/dist-packages (from transformers->peft) (0.23.1)

Requirement already satisfied: typer in /usr/local/lib/python3.13/dist-packages (from transformers->peft) (0.27.2)

Requirement already satisfied: cuda-pathfinder~=1.1 in
/usr/local/lib/python3.13/dist-packages (from cuda-
bindings<13,>=12.9.4->torch<3,>=2.4->bitsandbytes) (1.8.0)

Requirement already satisfied: anyio in
/usr/local/lib/python3.13/dist-packages (from httpx<1,>=0.23.0-
>huggingface_hub>=0.21.0->accelerate) (4.14.2)

Requirement already satisfied: certifi in
/usr/local/lib/python3.13/dist-packages (from httpx<1,>=0.23.0-
>huggingface_hub>=0.21.0->accelerate) (2026.7.22)

Requirement already satisfied: httpcore==1.* in
/usr/local/lib/python3.13/dist-packages (from httpx<1,>=0.23.0-
>huggingface_hub>=0.21.0->accelerate) (1.0.9)

Requirement already satisfied: idna in /usr/local/lib/python3.13/dist-
packages (from httpx<1,>=0.23.0->huggingface_hub>=0.21.0->accelerate)
(3.19)

Requirement already satisfied: h11>=0.16 in
/usr/local/lib/python3.13/dist-packages (from httpcore==1.*-
>httpx<1,>=0.23.0->huggingface_hub>=0.21.0->accelerate) (0.16.0)

Requirement already satisfied: mpmath<1.4,>=1.1.0 in
/usr/local/lib/python3.13/dist-packages (from sympy>=1.13.3-
>torch<3,>=2.4->bitsandbytes) (1.3.0)

Requirement already satisfied: MarkupSafe>=2.0 in
/usr/local/lib/python3.13/dist-packages (from jinja2->torch<3,>=2.4-
>bitsandbytes) (3.0.3)

Requirement already satisfied: shellingham>=1.3.0 in
/usr/local/lib/python3.13/dist-packages (from typer->transformers-
>peft) (1.5.4)

Requirement already satisfied: rich>=13.8.0 in
/usr/local/lib/python3.13/dist-packages (from typer->transformers-
>peft) (13.9.4)

Requirement already satisfied: annotated-doc>=0.0.2 in
/usr/local/lib/python3.13/dist-packages (from typer->transformers-
>peft) (0.0.5)

Requirement already satisfied: markdown-it-py>=2.2.0 in
/usr/local/lib/python3.13/dist-packages (from rich>=13.8.0->typer-
>transformers->peft) (4.2.0)

Requirement already satisfied: pygments<3.0.0,>=2.13.0 in
/usr/local/lib/python3.13/dist-packages (from rich>=13.8.0->typer-
>transformers->peft) (2.21.0)

Requirement already satisfied: mdurl~=0.1 in
/usr/local/lib/python3.13/dist-packages (from markdown-it-py>=2.2.0-
>rich>=13.8.0->typer->transformers->peft) (0.1.2)

```
{"model_id": "14751192d73a4e8c82673207414b3aaa", "version_major": 2, "version_minor": 0}
```

```
{"model_id": "371d3a1222d945b9ae25183e27931a52", "version_major": 2, "version_minor": 0}
```

```

{"model_id": "333db36e2ef04f61abd29faa2329562b", "version_major": 2, "version_minor": 0}

{"model_id": "07cbf824c04444988f36987d1e6aa61c", "version_major": 2, "version_minor": 0}

{"model_id": "ae27c868307646e6a0ef6daf3c722937", "version_major": 2, "version_minor": 0}

{"model_id": "acbe743f5b1f402babe7e518c5b3dc28", "version_major": 2, "version_minor": 0}

{"model_id": "06218bae998b4dbbb4c838b6d1ec6669", "version_major": 2, "version_minor": 0}

tokenizer = AutoTokenizer.from_pretrained("genaitraining/llama-2-7b-domain-tuned")
if tokenizer.pad_token is None:
    tokenizer.pad_token = tokenizer.eos_token # Llama 2 ships without a pad token

pipe = pipeline(
    "text-generation",
    model=model,
    tokenizer=tokenizer,
    max_new_tokens=256, # How long the generated answer can be
    return_full_text=False, # Return only the answer, not the prompt
    as well
)
llm = HuggingFacePipeline(pipeline=pipe)

# from_chain_type is the correct way to build this chain
qa_chain = RetrievalQA.from_chain_type(llm=llm, retriever=retriever)

query = "What are some effective strategies for managing and reducing stress-related symptoms, such as anxiety or insomnia?"
# .invoke replaces .run; the answer comes back under the "result" key
answer = qa_chain.invoke({"query": query})["result"]
print(answer)

{"model_id": "c60349762c8f4494a4a8350c2577a1de", "version_major": 2, "version_minor": 0}

{"model_id": "fe3264ec9a9b45e6a58a7d01eef7eded", "version_major": 2, "version_minor": 0}

{"model_id": "a3564292c4bb4873973741ced22ec889", "version_major": 2, "version_minor": 0}

```

[transformers] Passing `generation\_config` together with generation-related arguments=({'max_new_tokens'}) is deprecated and will be removed in future versions. Please pass either a `generation\_config`

object OR all generation parameters explicitly, but not both.
[transformers] Ignoring clean_up_tokenization_spaces=True for BPE tokenizer LlamaTokenizer. The clean_up_tokenization post-processing step is designed for WordPiece tokenizers and is destructive for BPE (it strips spaces before punctuation). Set clean_up_tokenization_spaces=False to suppress this warning, or set clean_up_tokenization_spaces_for_bpe_even_though_it_will_corrupt_output=True to force cleanup anyway.

The best approach depends on the individual. It's important to work with a mental health professional to identify the most effective therapy for your specific symptoms. Utilizing lifestyle changes such as limiting caffeine and alcohol consumption, getting regular exercise, keeping a diary, and practicing breathing exercises can help alleviate symptoms of stress and anxiety.

```
# ----- Evaluate the retriever: precision / recall / F1 / MRR @ k
-----
# There is no labelled test set for this data, so relevance is judged
by keywords:
# a chunk counts as relevant to a query if it contains any of that
query's keywords.
# EDIT EVAL_SET so the queries and keywords match what is actually in
RAG_info.txt.
```

```
EVAL_SET = [
    {
        "query": "How can I manage stress and maintain a healthy work-
life balance?",
        "keywords": ["stress", "work-life", "balance", "relax"],
    },
    {
        "query": "What should I eat to build muscle?",
        "keywords": ["protein", "muscle", "diet", "nutrition"],
    },
    {
        "query": "How much sleep do I need for recovery?",
        "keywords": ["sleep", "rest", "recovery"],
    },
    {
        "query": "How often should I do cardio?",
        "keywords": ["cardio", "aerobic", "running", "exercise"],
    },
]
```

```
K = 3 # how many chunks to retrieve per query
```

```
def is_relevant(text, keywords):
    """True if the chunk mentions any of the query's keywords."""
```

```

    text = text.lower()
    return any(kw.lower() in text for kw in keywords)

# A retriever that returns exactly K chunks, so precision@K is well
defined
eval_retriever = db.as_retriever(search_kwargs={"k": K})

results = []
for item in EVAL_SET:
    retrieved = eval_retriever.invoke(item["query"])

    # Relevant chunks among the ones we retrieved
    hits = [d for d in retrieved if is_relevant(d.page_content,
item["keywords"])]
    # Relevant chunks in the whole database - the denominator for
recall
    total_relevant = sum(1 for c in chunks if
is_relevant(c.page_content, item["keywords"]))

    precision = len(hits) / len(retrieved) if retrieved else 0.0
    # NOTE: recall is capped at K / total_relevant, so it stays low
when many
    # chunks mention the keywords. Compare runs at the same K rather
than to 1.0
    recall = len(hits) / total_relevant if total_relevant else 0.0
    f1 = 2 * precision * recall / (precision + recall) if (precision +
recall) else 0.0

    # Reciprocal rank: 1/position of the first relevant chunk (0 if
none found)
    rr = 0.0
    for rank, d in enumerate(retrieved, start=1):
        if is_relevant(d.page_content, item["keywords"]):
            rr = 1.0 / rank
            break

    results.append((item["query"], precision, recall, f1, rr,
total_relevant))

# Per-query table
print(f"Retrieval evaluation at k={K}\n")
print(f"{'query':<55} {'prec':>6} {'rec':>6} {'F1':>6} {'RR':>6}
{'#rel':>6}")
print("-" * 92)
for query, precision, recall, f1, rr, total_relevant in results:
    label = query if len(query) <= 52 else query[:49] + "..."
    print(f"{label:<55} {precision:>6.2f} {recall:>6.2f} {f1:>6.2f}
{rr:>6.2f} {total_relevant:>6}")

```

```

# Averages across all queries
n = len(results)
print("-" * 92)
print(f"Mean precision@{K}: {sum(r[1] for r in results) / n:.3f}")
print(f"Mean recall@{K}: {sum(r[2] for r in results) / n:.3f}")
print(f"Mean F1@{K}: {sum(r[3] for r in results) / n:.3f}")
print(f"MRR: {sum(r[4] for r in results) / n:.3f}")
# Hit rate = share of queries with at least one relevant chunk in the
top K
print(f"Hit rate@{K}: {sum(1 for r in results if r[4] > 0) /
n:.3f}")

```

Retrieval evaluation at k=3

query				prec	rec
F1	RR	#rel			

How can I manage stress and maintain a healthy wo...				1.00	0.19
0.32	1.00	16			
What should I eat to build muscle?				0.67	0.03
0.05	1.00	73			
How much sleep do I need for recovery?				1.00	0.18
0.30	1.00	17			
How often should I do cardio?				1.00	0.05
0.10	1.00	56			

Mean precision@3: 0.917					
Mean recall@3: 0.111					
Mean F1@3: 0.193					
MRR: 1.000					
Hit rate@3: 1.000					